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Conflict-Aware User Scheduling and Hybrid Beamforming for Near-Field STAR-RIS Systems

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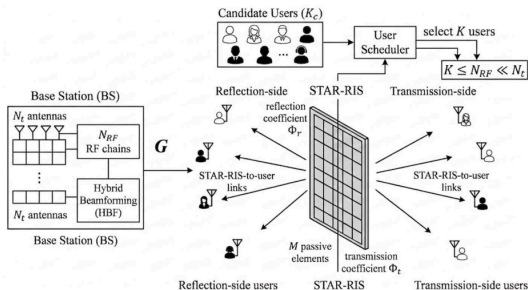
1. Context & Motivation

STAR-RIS: A simultaneously transmitting and reflecting surface that covers both sides of space. Users only receive; T-side or R-side is decided by location. In this work the STAR coefficients are already set for the slot.

Hybrid beamforming: The base station uses a limited number of RF chains, so only a few analog beams and a few users can be served in each slot. Each scheduled user is assigned one analog beam from a codebook, then a digital precoder.

Constraint: The candidate pool is larger than the stream budget. In the near field, users whose effective channels point the same way at the BS fight for the same analog beam. Top-gain collides; plain semi-orthogonal selection drops useful users.

2. Overall Architecture



STAR-RIS architecture

- **Limited hybrid stream budget**
Few RF chains, so only K users are served in one slot, and each of them gets one analog beam.
- **Conflict-aware user selection**
Rank users by strength, beam conflict, and orthogonality, then grow a set whose analog directions stay apart.
- **STAR coefficients are an input**
 Φ_T and Φ_R are preconfigured; the panel only applies those coefs. The BS outputs S and F , not a new surface.

3. System Model

Overview. A limited stream budget requires a small S . Effective channels $\tilde{h}_k$ already include the fixed STAR. Two users with a small angle between $\tilde{h}$ share analog beams and interfere. Scheduling keeps analog directions nearly orthogonal, then hybrid precoding separates the chosen set.

1. $\tilde{h}_k$ — BS view

Direct path plus STAR path with Φ_{s_k} already inside. How the BS sees user k .

2. ρ — correlation

$\rho_{A,B} = |\tilde{h}_A^H \tilde{h}_B| / (\|\tilde{h}_A\| \|\tilde{h}_B\|)$.
Small angle at the BS means high ρ and beam conflict.

3. η_k — priority

$\eta_k = \alpha_g g_k - \alpha_c c_k + \alpha_o \xi_k$.
Strength minus conflict plus orthogonality; not a PF score.

6. F_{BB} — digital

Built on $\{\tilde{h}_k : k \in S\}$.
Cleans leftover interference; cannot fix two users on one beam.

5. F_{RF} — analog

One codebook beam per user in S .
One analog beam $\leftrightarrow$ one user.

4. S — scheduled set

First user: $\max \eta$. Later: keep only semi-orthogonal survivors, then $\max \eta$. $|S| = K$.

4. Proposed CA-SUS-HBF Scheme

Approach. Rather than redesigning the STAR surface or searching all subsets, the six blocks run in one pass: score the pool, grow a semi-orthogonal S , assign codebook beams, then refine with a rate–fairness utility. Input: $K_C, \{\tilde{h}_k(\Phi_{s_k})\}, \mathcal{F}, K$. Output: $S, F_{\text{RF}}, F_{\text{BB}}$.

1. Scores

Compute $g_k, c_k = \sum_{j \in C_k} \omega_{k,j}$, and η_k .
Turns channels into quality, conflict, and orthogonality.

2. Reduced list

Sort by η and keep the most promising candidates.
Avoids scanning the whole pool at every later step.

3. Select S

First: $\max \eta$. Next: drop users too aligned with S , then $\max \eta$.
One user $\leftrightarrow$ one analog beam.

6. Refine

Local swaps if U rises; tail-repair if a user is below R_{th} .
Small search, not all subsets.

5. Utility U

Mix sum-rate, log-fairness, and a rate-floor penalty.
 α, λ_q from an offline validation sweep.

4. Hybrid F

Codebook F_{RF} for S , then digital F_{BB} .
Analog and baseband beams for the chosen users.

5. Simulation Results (Rate)

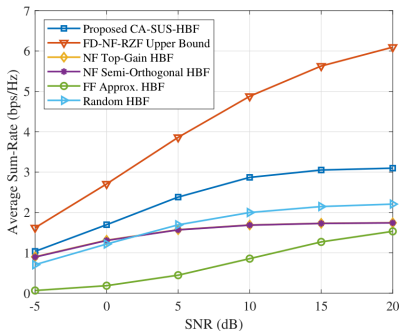


Figure 1. Average sum-rate versus SNR for different HBF schemes in the considered near-field STAR-RIS systems.

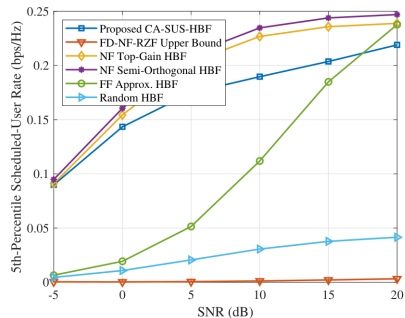


Figure 2. 5th-percentile scheduled-user rate versus SNR for different HBF schemes in the considered near-field STAR-RIS system.

6. Simulation Results (Users and Fairness)

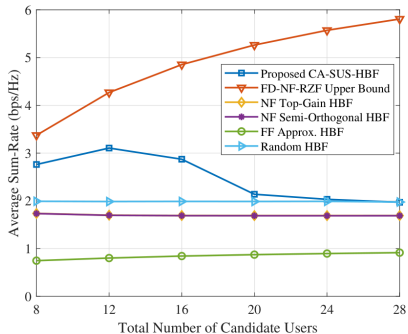


Figure 3. Average sum-rate versus the total number of candidate users.

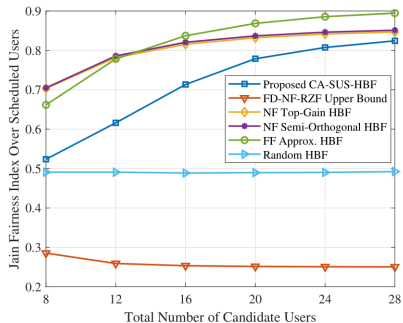


Figure 4. Jain's fairness index versus the total number of candidate users.

7. References

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Thank You

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